UIDAI DATA HACKATHON 2026

Data Driven Innovation on Aadhaar

INTRODUCTION

Project Name	State-wise Analysis of Aadhaar Biometric Usage, Enrolment Patterns and Demographic Updates
Aim	To analyze Aadhaar biometric, enrolment and demographic datasets in order to understand state-wise and age-wise usage patterns across India.
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DATASETS USED

The following datasets were used in this analysis:

- Biometric Dataset

Columns: date, state, district, pincode, bio_age_5_17, bio_age_17

- Enrolment Dataset

Columns: date, state, district, pincode, age_0_5, age_5_17, age_18_greater

- Demographic Dataset

Columns: date, state, district, pincode, demo_age_5_17, demo_age_17

OBJECTIVES

- To clean and merge large-scale Aadhaar datasets.
- To perform state-wise and age-wise analysis of enrolment, biometric, and demographic data.
- To identify key trends and anomalies across regions.
- To visualize insights using graphs for easier interpretation.

METHODOLOGY

The analysis followed a structured pipeline consisting of the following steps:

- Data Loading: Multiple CSV files were merged for each dataset.
- Data Cleaning: Removal of empty rows and duplicate records.
- Feature Engineering: Creation of total usage metrics.
- Analysis State-wise aggregation and ranking.
- Visualization: Graphs generated to represent insights.

ANALYSIS AND RESULT

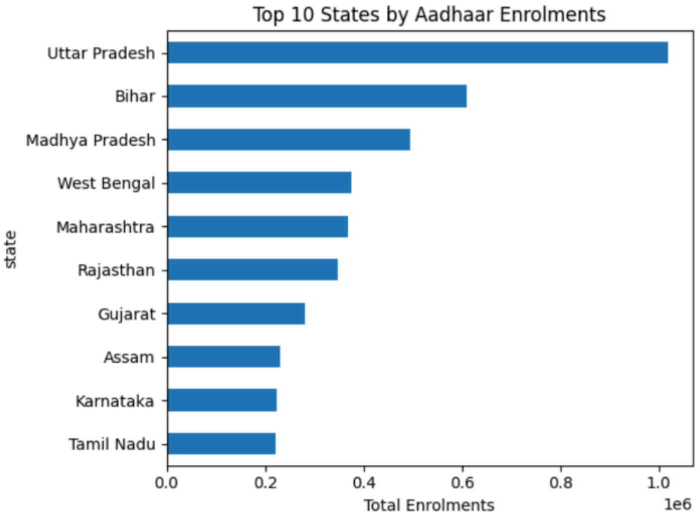
Aadhaar Enrolment Analysis: Trends, Anomalies & Visuals

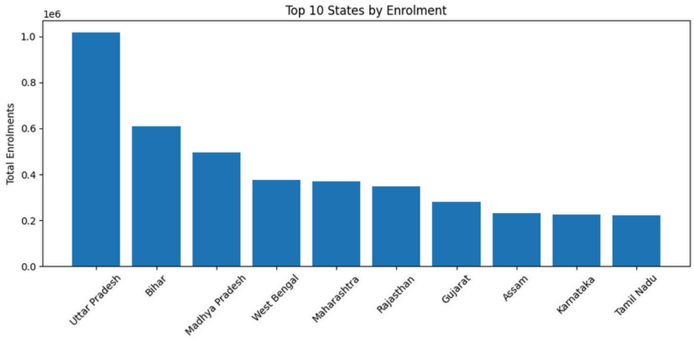
Key Trends and Anomalies

- Enrolment data is highly concentrated in a few large states, indicating population driven patterns.
- Uttar Pradesh records the highest Aadhaar enrolments, followed by Bihar and Madhya Pradesh.
- The majority of new enrolments belong to the 0–17 age group, showing adult Aadhaar saturation.
- Adult enrolments are comparatively low, suggesting Aadhaar maturity among adults.
- Northern and Central India show higher enrolment activity compared to other regions.
- Union Territories and island regions show very low enrolment counts.
- Uttar Pradesh appears as a statistical outlier due to extremely high enrolment volume.
- An invalid state entry '100000' is present, indicating a possible data mapping error.
- Duplicate naming of Andaman & Nicobar Islands highlights data consistency issues.

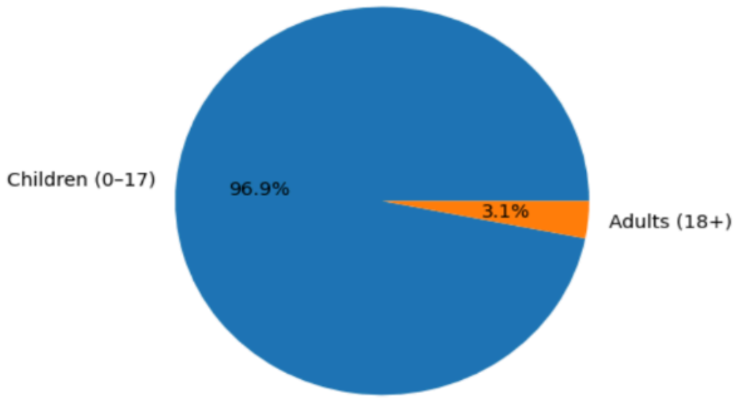
Visual Analysis

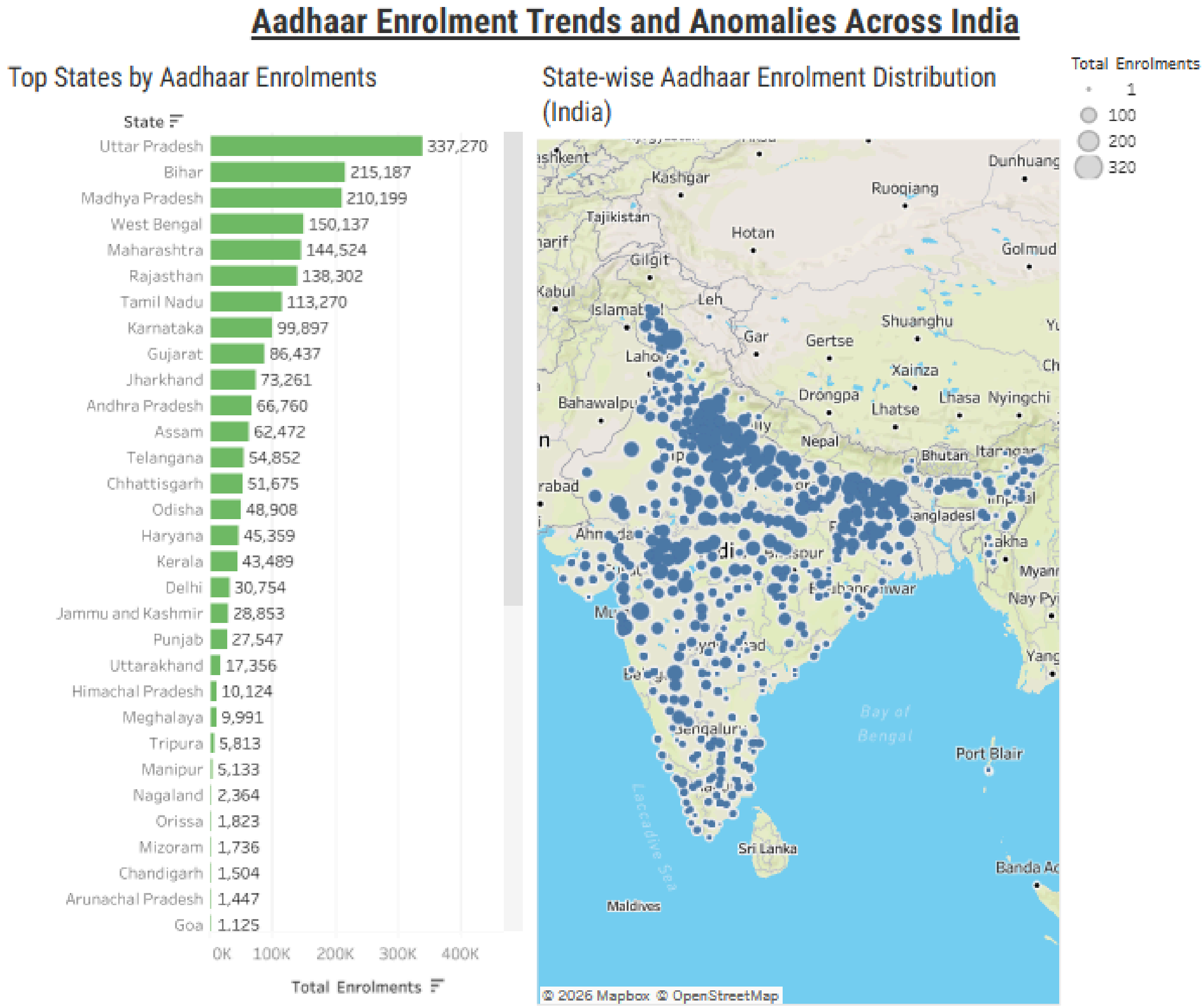
Top 10 States by Aadhaar Enrolments:





Overall Age-wise Aadhaar Enrolment Distribution:





These patterns suggest that Aadhaar enrolment efforts are now largely focused on younger populations, while adult enrolment has reached saturation in most states.

Demographic Update Trends and Anomalies Report

This report analyzes Aadhaar demographic update data across India. Biometric records were excluded, and only demographic update datasets were considered. The top 10 states were selected based on total demographic update volume.

Key Trends

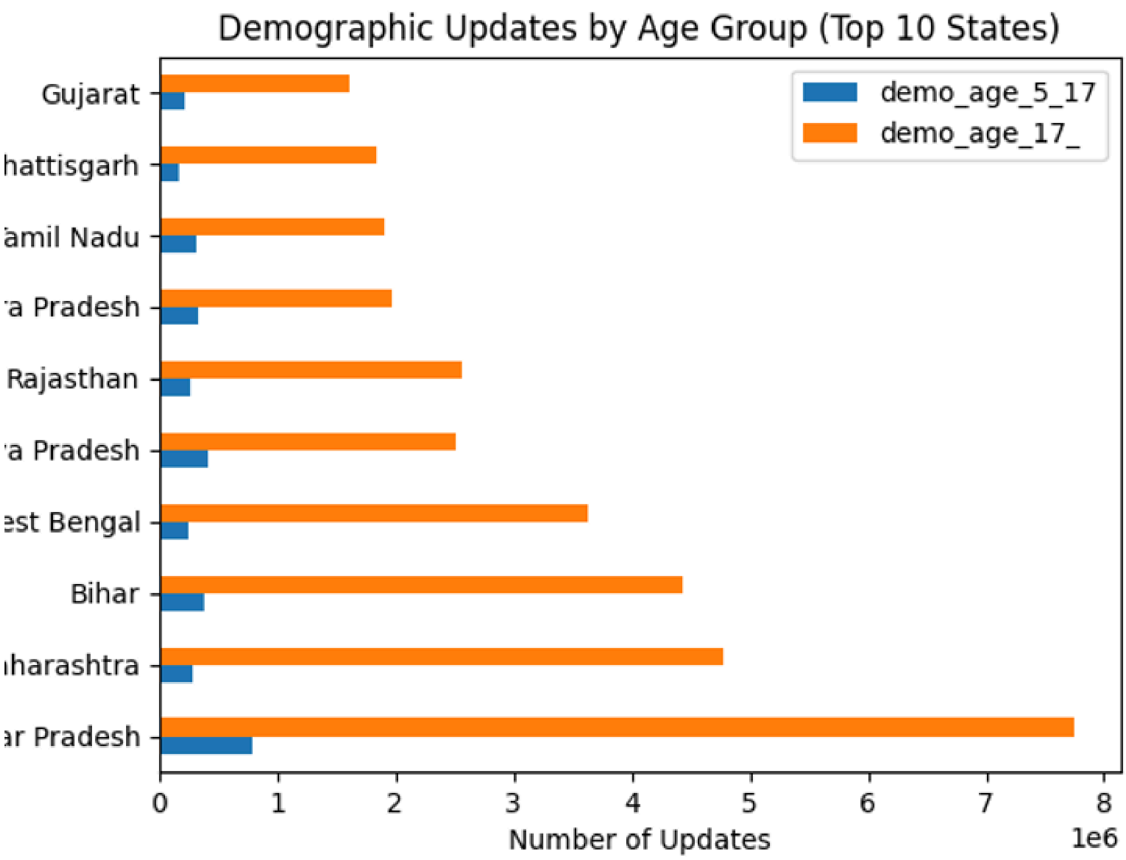
- Demographic updates are heavily concentrated in highly populated states.
- A significantly higher number of updates are observed in the 17+ age group, indicating corrections during adulthood such as name, address, and date-of-birth updates.
- Updates in the 5–17 age group are comparatively lower and primarily driven by school-age corrections.

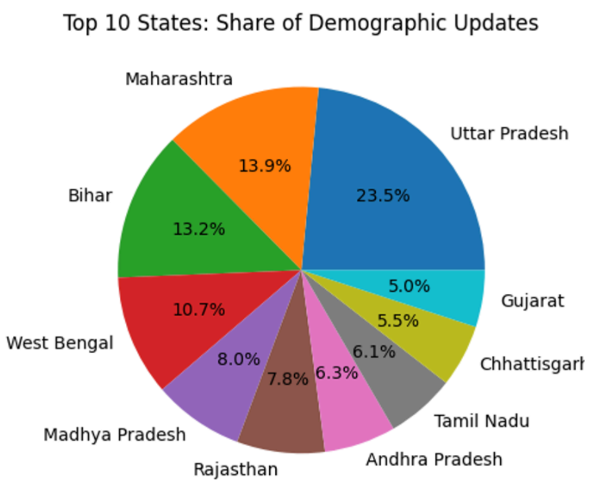
Anomalies Observed

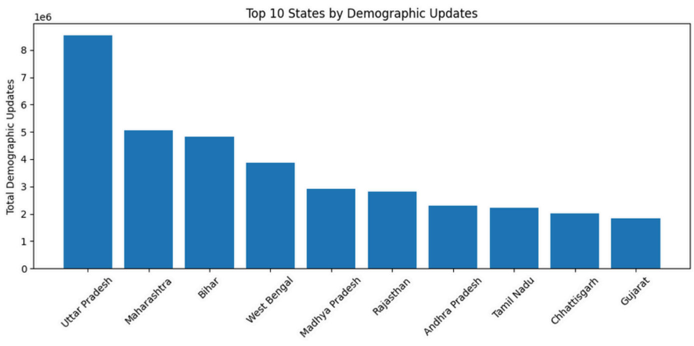
- Certain states show unusually high adult demographic updates relative to child updates, suggesting mass correction drives or policy-led cleanup exercises.

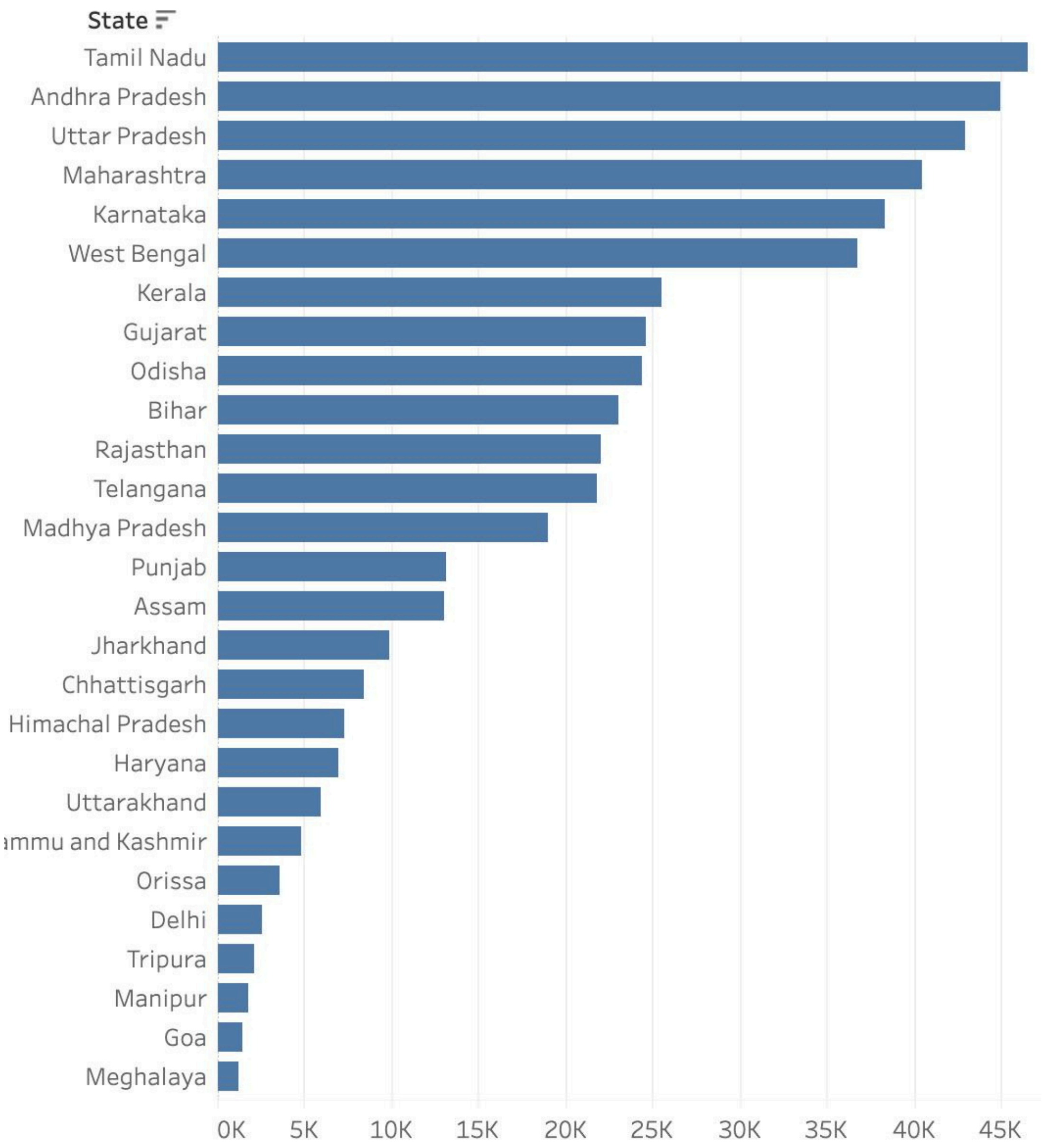
- Some demographically large states appear below expectations, possibly due to digital access gaps or underreporting at enrollment centers.

Visual Analysis









The findings indicate that demographic update behavior is strongly age dependent and policy driven. These insights can help government to optimize outreach strategies and improve update compliance across states.

Aadhaar Biometric Data Trends and Anomalies Report

Dataset Overview

This dataset contains aggregated Aadhaar biometric activity across India, segmented by date, state, district, pincode, and age groups (5–17 years and 17+ years). It reflects biometric updates and authentication usage patterns.

Key Trends

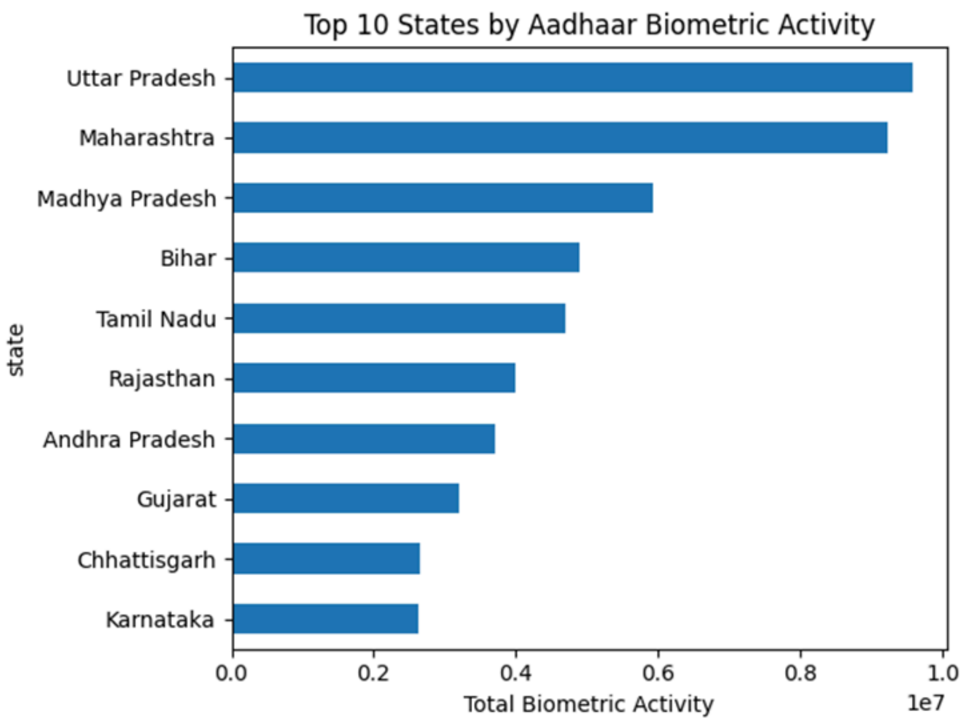
- Biometric activity is significantly higher in the 17+ age group across most regions.
- Large and populous states dominate biometric usage.
- Considerable variation exists at the district level within states.
- Repeated biometric activity across dates suggests ongoing Aadhaar usage.

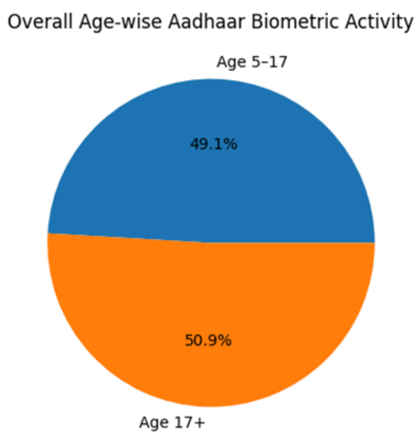
Anomalies Observed

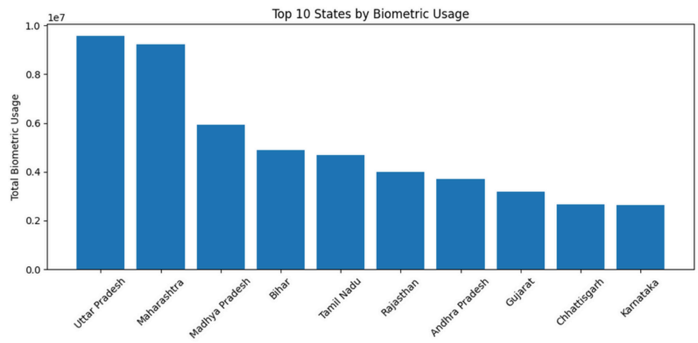
- Certain districts and states show extremely high biometric counts, acting as usage hotspots.
- Some regions report very low biometric activity, possibly due to stabilized enrolments or access limitations.
- Strong imbalance between age groups in some regions highlights adult-driven biometric demand.

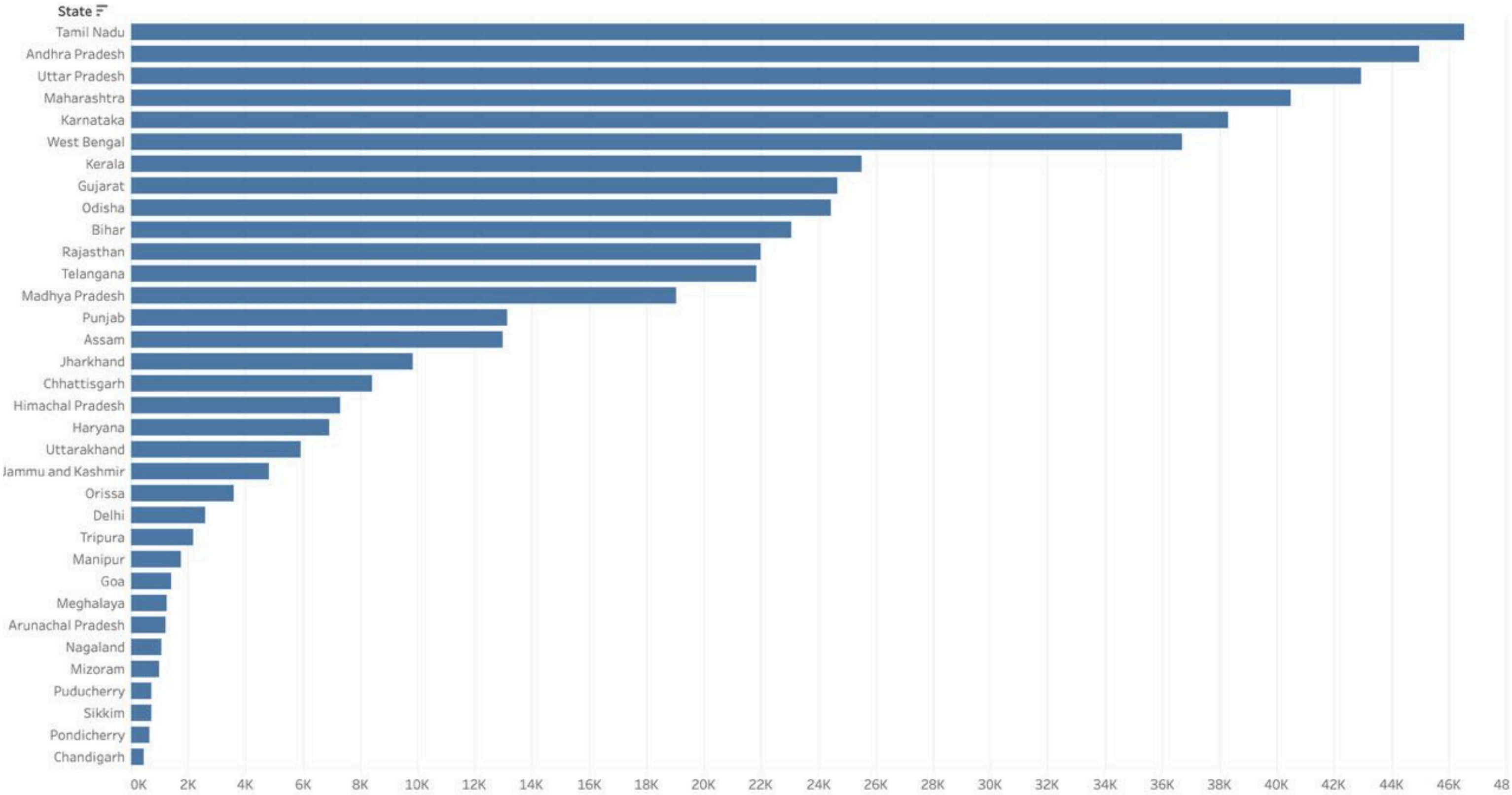
Visual Analysis

Top 10 States by Aadhaar Biometric Activity:









The biometric analysis reveals strong state-wise variation in Aadhaar biometric usage, highlighting differences in system adoption and frequency of authentication across regions.

KEY INSIGHTS

- Biometric usage shows strong state-wise variation.
- Aadhaar enrolment activity differs significantly across regions.
- Demographic updates are higher in states with greater population movement.
- Adult age groups contribute the most to biometric usage.
- Graphical analysis helps clearly identify usage trends.

CONCLUSION

The analysis of Aadhaar datasets provided clear insights into state-wise and age-wise usage patterns of biometric, enrolment, and demographic services. The results show that Aadhaar service usage varies significantly across regions, and data visualization played an important role in understanding these trends.

CODE APPENDIX

The following code snippets were written in Python using Pandas and Matplotlib libraries.

Data Loading Function

Data Cleaning Function

```
#------------
#LOADING DATA
#------------
def load_merge(path,file):
    print("\nlOADING",file)
    files = glob.glob(path)
    dataframes = []
    for f in files:
        print('READING:',f)
        df = pd.read_csv(f)
        dataframes.append(df)

    if len(dataframes) == 0:
        print('NO FILES FOUND')
        return None

    merge_df = pd.concat(dataframes,ignore_index=True)
    print('LOADED',file,'SHAPE',merge_df.shape)
    return merge_df
```

```
#---------------
#CLEANING FILTER
#---------------
def clean_data(data,file):
    print(f"\nCLEANING {file} DATA")
    before = data.shape[0]
    df = data.dropna(how = 'all')
    after = data.shape[0]
    print('REMOVE EMPTY ROWS',before - after)

    before = data.shape[0]
    df = data.drop_duplicates()
    after = data.shape[0]
    print('REMOVE DUPLICATE ROWS',before - after)
    return data
```

Data Analysis

```
#------------------
#ANALYSING THE DATA
#------------------
def create_total_column(df, columns, new_col_name):
    #Adds a new column by summing given columns
    df[new_col_name] = df[columns].sum(axis=1)
    return df


def state_wise_analysis(df, value_col, label):
    #Groups data by state and sums the value column
    print(f"\nSTATE WISE ANALYSIS FOR {label}")
    state_data = (df.groupby('state')[value_col].sum().sort_values(ascending=False))
    print("\nTop 5 States:")
    print(state_data.head(5))
    print("\nBottom 5 States:")
    print(state_data.tail(5))
    return state_data
```

```
print("\n-----ENROLMENT ANALYSIS-----")
# Create total enrolment count
enrolment_df = create_total_column(enrolment_df,[
                'age_0_5', 'age_5_17',
                'age_18_greater'],
                'total_enrolments')
# Quick check
print(enrolment_df[[
     'age_0_5', 'age_5_17',
     'age_18_greater',
     'total_enrolments']].head())
# State-wise enrolment analysis
enrolment_state_usage = state_wise_analysis(enrolment_df,
                         'total_enrolments','ENROLMENT')
```

```
print("\n-----DEMOGRAPHIC ANALYSIS-----")
# Create total demographic updates
demographic_df = create_total_column(demographic_df,[
                 'demo_age_5_17', 'demo_age_17_'],
                 'total_demographic_updates')
# Quick check
print(demographic_df[['demo_age_5_17',
                      'demo_age_17_',
                      'total_demographic_updates']].head())
# State-wise demographic updates
demographic_state_usage = state_wise_analysis(demographic_df,
                          'total_demographic_updates',
                          'DEMOGRAPHIC UPDATES')
```

```
print("\n-----BIOMETRIC ANALYSIS-----")
# Create total biometric usage
biometric_df = create_total_column(biometric_df,[
               'bio_age_5_17', 'bio_age_17_'],
               'total_biometric_usage')
# Quick check
print(biometric_df[['bio_age_5_17',
                    'bio_age_17_',
                    'total_biometric_usage']].head())
# State-wise biometric usage
biometric_state_usage = state_wise_analysis(biometric_df,
                        'total_biometric_usage','BIOMETRIC USAGE')
```

Graphical Representation

```
#------------------------
#GRAPHICAL REPRESENTATION
#------------------------
def graph(s_series,title,lable,nfile):
    mlp.figure(figsize=(10,5))
    states=s_series.head(10)
    mlp.bar(states.index, states.values)
    mlp.xticks(rotation=45)
    mlp.ylabel(lable)
    mlp.title(title)
    mlp.tight_layout()

    mlp.savefig(nfile)
    mlp.close()

    print(f"Saved graph: {nfile}")
```

```
graph(enrolment_state_usage,"Top 10 States by Enrolment","Total Enrolments","enrolment_by_state.png")
graph(demographic_state_usage,"Top 10 States by Demographic Updates","Total Demographic Updates","demographic_updates_by_state.png")
graph(biometric_state_usage,"Top 10 States by Biometric Usage","Total Biometric Usage","biometric_usage_by_state.png")
```